

Algebra II with Trigonometry

Chapter 5.1

Olympiad Problems

Saved tutoring response with MathJax rendering. Original PDF file:

[4401267_alg-2trig-h-chp-5-1-note-docx.pdf](#)

Problems

1. Suppose a circle of radius r has a sector with central angle θ (in radians) whose area equals the arc length of the same sector. Find, in terms of r , all positive values of θ that satisfy this property.
2. How many distinct positive integers n are there such that n° and $2n^\circ$ are coterminal angles in standard position?
3. The minute and hour hand of a clock are both 10 cm long. At exactly 12:00, both hands are superimposed. After one hour, what is the exact total area traced out by the tips of the hands, treating each path as a sector swept on a circle of radius 10 cm? Express your answer in terms of π .
4. Let A and B be two angles (in degrees), both less than 360° , such that the arc lengths subtended by A in a circle of radius r and by B in a circle of radius $2r$ are equal. If $A : B = 3 : 1$, what is the value of A ?
5. A Ferris wheel with radius 15 m completes one full counterclockwise revolution every 40 seconds. Another identical Ferris wheel next to it rotates in the opposite direction, making exactly n revolutions during the same amount of time. If the points at the topmost positions of the wheels align every 10 seconds, what is the value of n ?

6. The area of a sector of a circle is numerically equal (in the same units) to five times the circumference of the circle. Find the central angle of the sector in radians, in terms of r .